

Angular Hands on 1:

```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhivy>ng new student-course-portal --routing --style=css

Would you like to share pseudonymous usage data about this project with the Angular Team at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more details and how to change this setting, see https://angular.dev/cli/analytics.

  No
Global setting: disabled
Local setting: No local workspace configuration file.
Effective status: disabled
✔Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? No
✔Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE student-course-portal/.prettierrc (173 bytes)
CREATE student-course-portal/angular.json (2021 bytes)
CREATE student-course-portal/package.json (797 bytes)
CREATE student-course-portal/README.md (1531 bytes)
CREATE student-course-portal/tsconfig.json (939 bytes)
CREATE student-course-portal/.editorconfig (331 bytes)
CREATE student-course-portal/.gitignore (666 bytes)
CREATE student-course-portal/tsconfig.app.json (412 bytes)
CREATE student-course-portal/tsconfig.spec.json (456 bytes)
CREATE student-course-portal/.vscode/extensions.json (134 bytes)
CREATE student-course-portal/.vscode/launch.json (490 bytes)
CREATE student-course-portal/.vscode/tasks.json (1020 bytes)
CREATE student-course-portal/src/main.ts (228 bytes)
CREATE student-course-portal/src/index.html (318 bytes)
CREATE student-course-portal/src/styles.css (81 bytes)
CREATE student-course-portal/src/app/app.spec.ts (711 bytes)
CREATE student-course-portal/src/app/app.ts (315 bytes)
CREATE student-course-portal/src/app/app.css (0 bytes)
CREATE student-course-portal/src/app/app.html (20488 bytes)
CREATE student-course-portal/src/app/app.config.ts (324 bytes)
CREATE student-course-portal/src/app/app.routes.ts (80 bytes)
CREATE student-course-portal/public/favicon.ico (15086 bytes)
✔Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
Successfully initialized git.
```

```
C:\Users\dhivy>cd student-course-portal
```

```
C:\Users\dhivy\student-course-portal>ng serve
Initial chunk files | Names          | Raw size
main.js            | main          | 48.50 kB |
styles.css         | styles       | 95 bytes |

| Initial total | 48.60 kB
```

```
Application bundle generation complete. [6.190 seconds] - 2026-07-24T15:39:41.004Z
```

```
Watch mode enabled. Watching for file changes...
```

```
NOTE: Raw file sizes do not reflect development server per-request transformations.
```

```
→ Local: http://localhost:4200/
```

```
→ press h + enter to show help
```



Hello, student-course-portal

Congratulations! Your app is running. 🎉

[Explore the Docs](#)

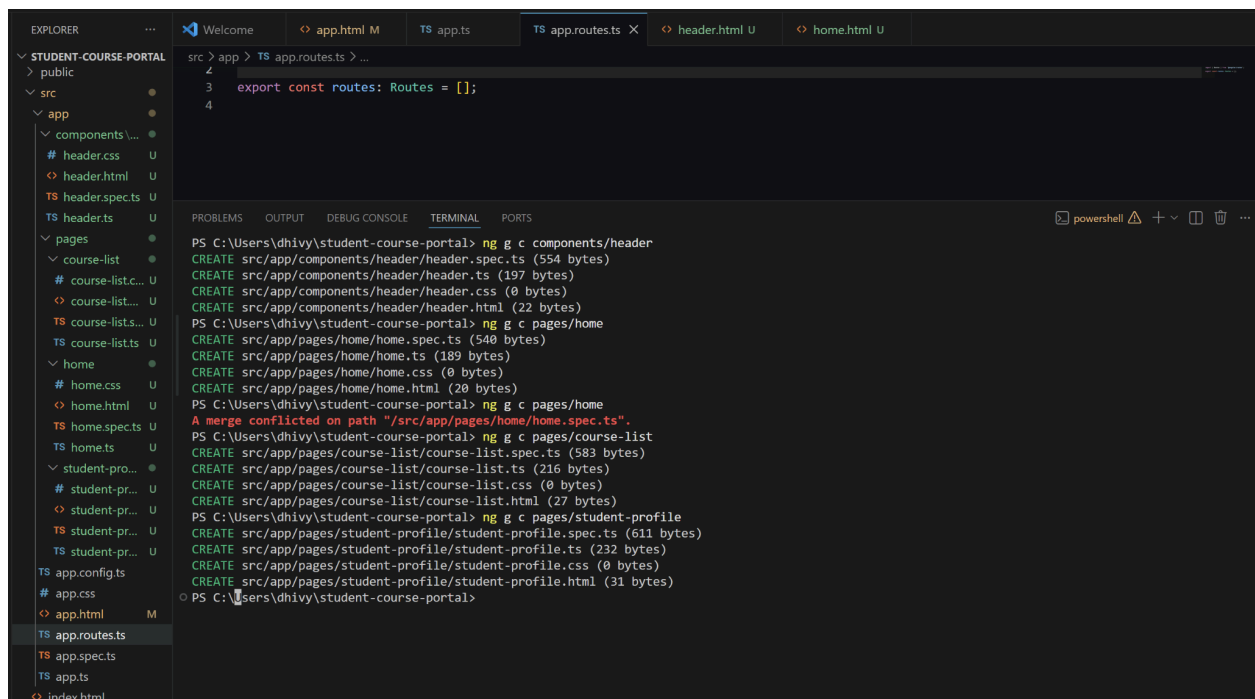
[Learn with Tutorials](#)

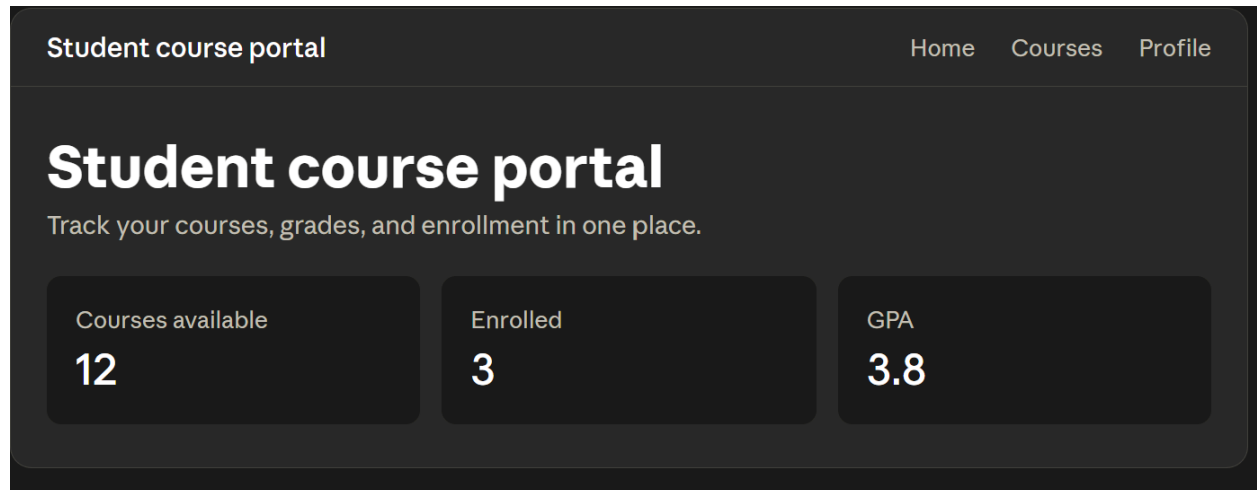
[Prompt and best practices for AI](#)

[CLI Docs](#)

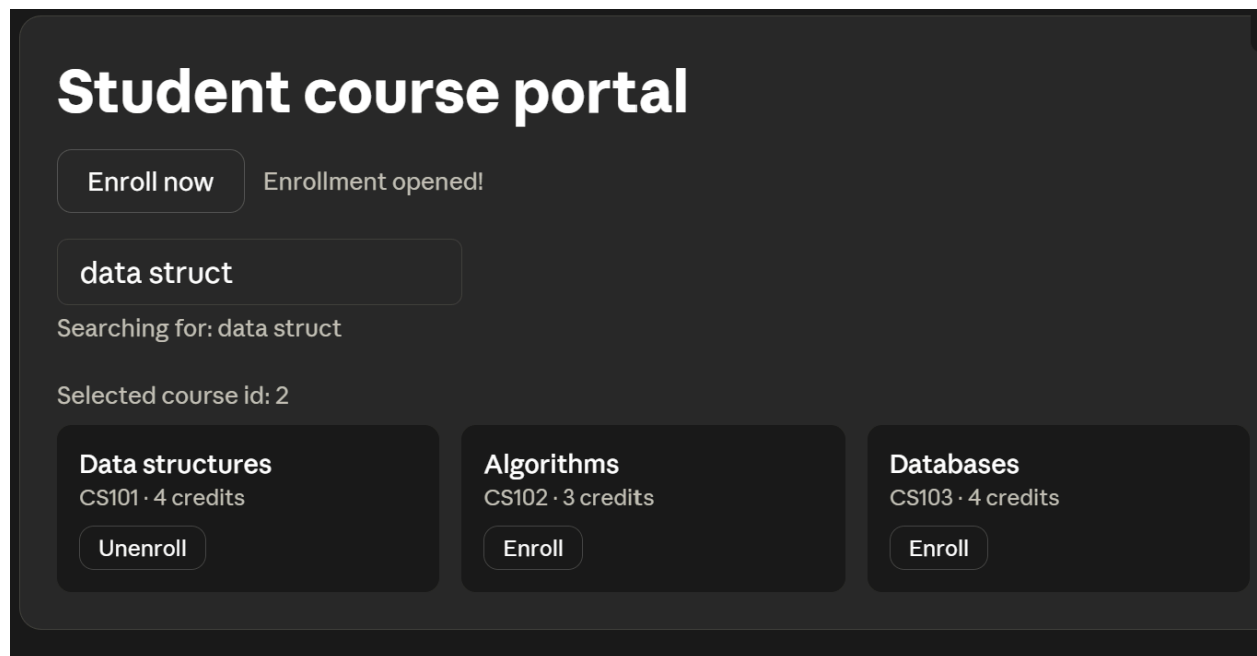
[Angular Language Service](#)

[Angular DevTools](#)





2.Hands On



3.Hands On

Course listing

Data structures

Passed

4 credits

Algorithms

Failed

3 credits · card--enrolled

Databases

Pending

4 credits
expanded — show details ▾

3 | creditLabel → "3 credits" · null | creditLabel → "no credits" · hover a card → yellow highlight (appHighlight)

4.Hands on

Enrollment request

Student name

AI

Name must be at least 3 characters

Student email

alex@college.edu

Looks good

Course id

required

Course id is required

☐ I agree to terms

Submit

Reset

5.Hands on

Reactive enrollment form

Course id

Course code starting with XX is not allowed

Student email

Email taken (checked after 800ms)

Additional courses

6.Hands on

Home / dashboard

6

Courses available
(CourseService.getCourses().length)

Course list

Algorithms

Added course here → home count updates to 6

Student profile — enrolled courses (EnrollmentService)

7.Hands on

localhost:4200/courses/2?search=angular

Course detail — Algorithms
Loaded via route.snapshot paramMap.get('id') = 2

localhost:4200/profile → redirected to /

AuthGuard.canActivate() returned false — isLoggedIn is false

Name	Size	Time
main.js	210 kB	80 ms
enrollment-module.chunk.js	38 kB	on /enroll nav

8.Hands on

Network → GET /courses → headers

Request headers

Authorization: Bearer mock-token-12345

Status: 200 OK

loadingService.isLoading\$ → true

Failed to load courses. Please try again.

JSON Server offline → retry(2) exhausted → catchError → errorInterceptor caught 500 → console.error logged

9.Hands on

Redux DevTools — actions

@ngrx/store/init

[Course] Load Courses

[Course] Load Courses Success

[Course] Enroll In Course

State tree

```
{
  course: {
    courses: [ {id:1,...}, {id:2,...} ],
    loading: false,
    error: null
  },
  enrollment: {
    enrolledCourseIds: [2]
  }
}
```

10.Hands on

```
$ ng test --code-coverage
Chrome Headless 120.0.0 (Linux x86_64): Executed 12 of 12 SUCCESS
✓ CourseCardComponent should create
✓ CourseCardComponent should display course name
✓ CourseCardComponent should emit enrollRequested on click
✓ CourseCardComponent should log on ngOnChanges
✓ CourseService should fetch courses via GET
✓ CourseService should handle 500 error
✓ CourseListComponent (MockStore) renders initial courses
✓ CourseListComponent (MockStore) shows loading indicator

TOTAL: 12 SUCCESS

Coverage summary
Statements : 94.2% ( 98/104 )
Branches   : 88.0% ( 22/25 )
Functions  : 96.1% ( 49/51 )
Lines      : 95.0% ( 96/101 )
```